

ML Practical List

1. The advertising dataset captures the sales revenue generated with respect to advertisement costs across multiple channels like Radio, TV, and Newspapers.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the sales w.r.t Radio features.
- iii. Also evaluate the model using scores RMSE

2. The advertising dataset captures the sales revenue generated with respect to advertisement costs across multiple channels like radio, tv, and newspapers.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the sales w.r.t attribute tv.
- iii. Also evaluate the model using scores RMSE

3. The advertising dataset captures the sales revenue generated with respect to advertisement costs across multiple channels like radio, tv, and newspapers.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the sales w.r.t attribute newspaper.
- iii. Also evaluate the model using scores RMSE

4. The advertising dataset captures the sales revenue generated with respect to advertisement costs across multiple channels like radio, tv, and newspapers.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the sales w.r.t Radio and TV
- iii. Also evaluate the model using scores RMSE

5. The advertising dataset captures the sales revenue generated with respect to advertisement costs across multiple channels like radio, tv, and newspapers.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the sales w.r.t Newspaper and TV
- iii. Also evaluate the model using scores RMSE

6. The advertising dataset captures the sales revenue generated with respect to advertisement costs across multiple channels like radio, tv, and newspapers.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the sales w.r.t Newspaper and Radio
- iii. Also evaluate the model using scores RMSE

7. The car dataset captures the selling price of the used cars with respect to features like year_bought, km_driven, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t year brought
- iii. Also evaluate the model using scores RMSE

8. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t `km_driven`
- iii. Also evaluate the model using scores RMSE

9. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t transmission
- iii. Also evaluate the model using scores RMSE

10. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t owner
- iii. Also evaluate the model using scores RMSE

11. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t year brought and `km_driven`
- iii. Also evaluate the model using scores RMSE

12. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t year brought and transmission
- iii. Also evaluate the model using scores RMSE

13. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t year brought and owner
- iii. Also evaluate the model using scores RMSE

14. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression model to predict the selling prices w.r.t year brought and owner
- iii. Also evaluate the model using scores RMSE

15. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, transmission and owner.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Build Regression models to predict the selling prices w.r.t km driven and transmission
- iii. Also evaluate the model using scores RMSE

16. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, `transmission` and `owner`.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Build Regression models to predict the selling prices w.r.t km driven and owner
 - Also evaluate the model using scores RMSE
17. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, `transmission` and `owner`.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Build Regression models to predict the selling prices w.r.t transmission and owner
 - Also evaluate the model using scores RMSE
18. The car dataset captures the selling price of the used cars with respect to features like `year_bought`, `km_driven`, `transmission` and `owner`.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Build Regression models to predict the selling prices w.r.t transmission and owner
 - Also evaluate the model using scores RMSE
19. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Use a SVM to classify whether a user purchased a car or not? (**Use Linear Kernel**)
 - Create a confusion matrix and evaluate the model using accuracy
20. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Use a SVM to classify whether a user purchased a car or not? (**Use Linear Kernel**)
 - Create a confusion matrix and evaluate the model using Recall.
21. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Use a SVM to classify whether a user purchased a car or not? (**Use Linear Kernel**)
 - Create a confusion matrix and evaluate the model using Precision
22. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Use a SVM to classify whether a user purchased a car or not? (**Use RBF Kernel**)
 - Create a confusion matrix and evaluate the model using F1-Measure
23. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.
Objectives:
- Understand the Dataset & cleanup (if required).
 - Use a SVM to classify whether a user purchased a car or not? (**Use RBF Kernel**)
 - Create a confusion matrix and evaluate the model using accuracy

24. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to classify whether a user purchased a car or not? **(Use RBF Kernel)**
- iii. Create a confusion matrix and evaluate the model using Recall.

25. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to classify whether a user purchased a car or not? **(Use RBF Kernel)**
- iii. Create a confusion matrix and evaluate the model using Precision

26. A social network dataset is a categorical dataset to determine whether a user purchased a particular product based on gender, age and estimated salary.

Objectives:

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to classify whether a user purchased a car or not? **(Use RBF Kernel)**
- iii. Create a confusion matrix and evaluate the model using F1-Measure

27. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to predict the iris plant category? **(Use Linear Kernel)**
- iii. Create a confusion matrix and evaluate the model using Precision.

28. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to predict the iris plant category? **(Use Linear Kernel)**
- iii. Create a confusion matrix and evaluate the model using Recall.

29. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to predict the iris plant category? **(Use Linear Kernel)**
- iii. Create a confusion matrix and evaluate the model using Accuracy.

30. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to predict the iris plant category? **(Use Linear Kernel)**
- iii. Create a confusion matrix and evaluate the model using F1-Measure.

31. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to predict the iris plant category? **(Use RBF Kernel)**
- iii. Create a confusion matrix and evaluate the model using Precision.

32. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

- i. Understand the Dataset & cleanup (if required).
- ii. Use a SVM to predict the iris plant category? **(Use RBF Kernel)**

- iii. Create a confusion matrix and evaluate the model using Recall.
33. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.
- i. Understand the Dataset & cleanup (if required).
 - ii. Use a SVM to predict the iris plant category? **(Use RBF Kernel)**
 - iii. Create a confusion matrix and evaluate the model using Accuracy.
34. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.
- i. Understand the Dataset & cleanup (if required).
 - ii. Use a SVM to predict the iris plant category? **(Use RBF Kernel)**
 - iii. Create a confusion matrix and evaluate the model using F1-Measure.
35. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a decision tree classifier to predict the iris plant category? **(Use GINI INDEX criteria, use max_depth=4, min_samples_split=2)**
 - iii. Evaluate the model using Accuracy.
36. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a decision tree classifier to predict the iris plant category? **(Use Entropy criteria, use max_depth=4, min_samples_split=2)**
 - iii. Evaluate the model using Accuracy.
37. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a decision tree classifier to predict the iris plant category? **(Use log loss criteria, use max_depth=4, min_samples_split=2)**
 - iii. Evaluate the model using Accuracy.
38. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a logistic Regression classifier to predict the iris plant category?
 - iii. Evaluate the model using Accuracy.
39. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a **Bagging Classifier** model to predict the iris plant category?
 - iii. Evaluate the model using Accuracy.
40. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a **Random Forest Classifier** model to predict the iris plant category?
 - iii. Evaluate the model using Accuracy.
41. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a **Gradient Boost Classifier** model to predict the iris plant category?
 - iii. Evaluate the model using Accuracy.
42. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build a **AdaBoost Classifier** model to predict the iris plant category?

- iii. Evaluate the model using Accuracy
43. The Iris data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. Apply Principal Component Analysis on the Iris dataset.
- i. Understand the Dataset & cleanup (if required).
 - ii. Build any one classifier of your choice on the dataset to predict the Iris plant category. (Without Applying PCA)
 - iii. Apply **PCA** technique for dimensionality reduction. And build the same classifier that you have chosen in step ii to predict the Iris plant category. Compare the results.